MOBILE SALES PERFORMANCE DASHBOARD

Business Analysis Report · Power BI · DAX · SQL · Data Modelling

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github.com/Ajay221301/Mobile_sales_dash | May 2026

Dataset: 3,835 Transactions · 4 Years (2021–2024) · 19 Cities · 5 Brands

1. Executive Summary

This report presents a comprehensive business intelligence analysis of mobile phone sales across **19 cities, 5 brands, and 4 years (2021–2024)** using a dataset of 3,835 transactions. Built end-to-end — from raw Excel data through SQL business queries and Power Query transformation to an interactive 3-page Power BI dashboard with DAX time-intelligence measures.

The project answers six core business questions: which cities and brands drive revenue, how sales trend over time (MTD/YoY), what payment methods customers prefer, how satisfaction varies, and where growth is declining.

TOTAL REVENUE	TRANSACTIONS	UNITS SOLD	AVG PRICE/UNIT
■76.9 Cr	**3,835**	**19,150**	**■40,114**
2021–2024	4 Years	5 Brands	Per Transaction

Year-on-Year Performance Snapshot

Year	Total Revenue	Transactions	YoY Growth	Signal
2021	■5.88 Cr	281	—	Base Year
2022	■26.20 Cr	1,250	**+345.4%**	↑ Peak Growth
2023	■25.33 Cr	1,249	**–3.3%**	↓ Slight Dip
2024	■19.51 Cr	1,055	**–23.0%**	↓ Declining

2. Brand & Geographic Revenue Analysis

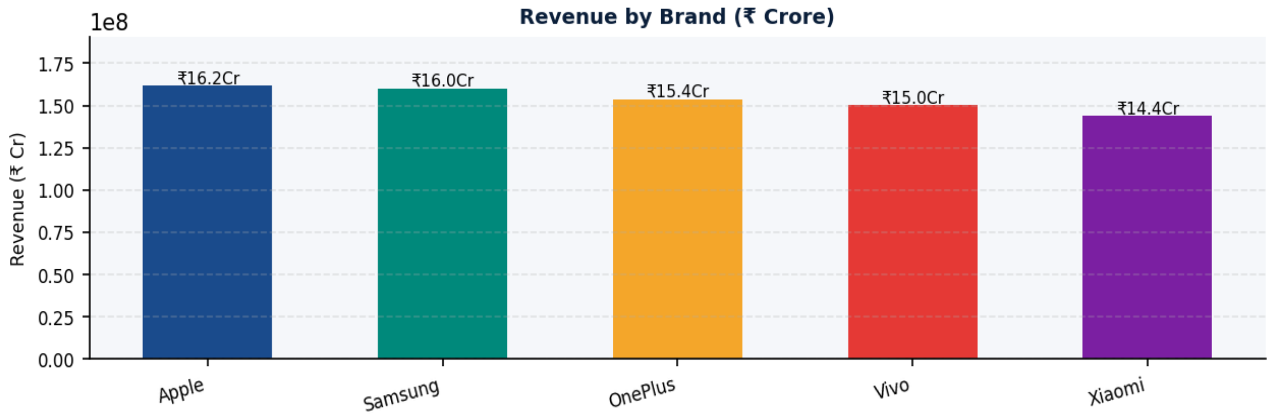


Fig 1 — Apple leads at ■16.2 Cr, with Samsung close behind at ■16.0 Cr. All 5 brands compete within a tight ±12% revenue band, indicating no dominant player.

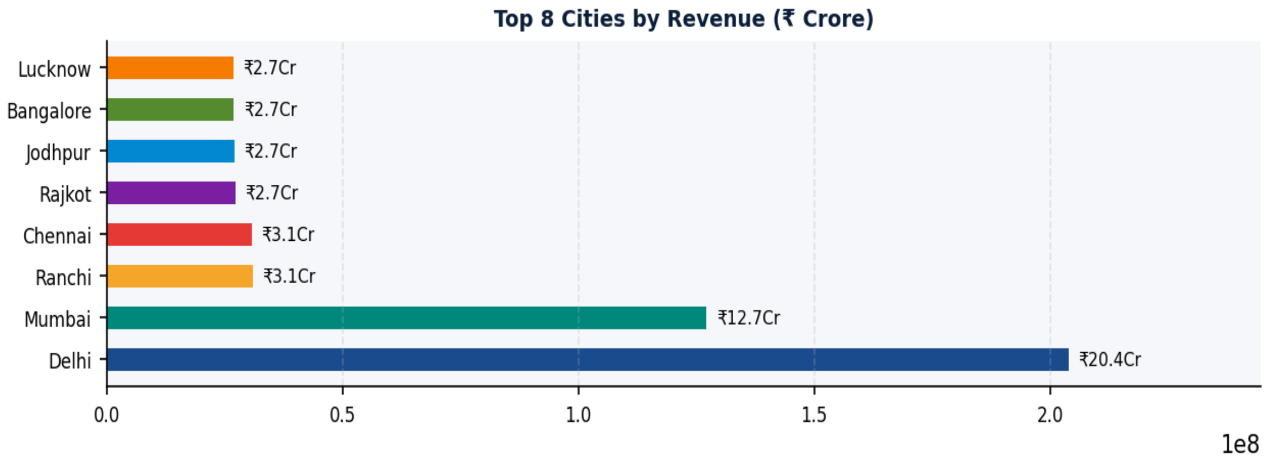
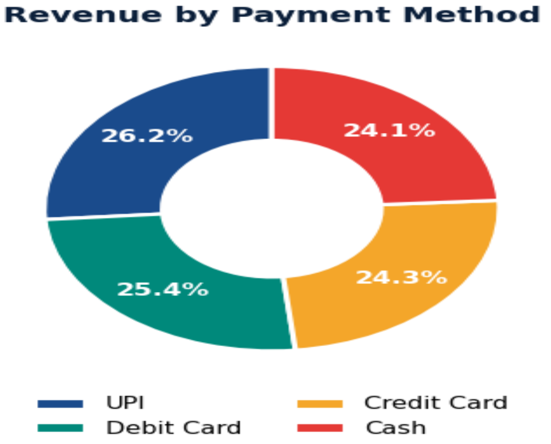


Fig 2 — Delhi dominates with ■20.4 Cr (26.5% market share). Mumbai second at ■12.7 Cr (16.5%). Together these two cities account for 43% of total revenue.

Apple Leads Revenue
■16.2 Cr. Highest avg unit price ~■55K+. Premium segment drives top-line.

Delhi = 26.5% Share
■20.4 Cr from Delhi alone — nearly 3x Mumbai. #1 priority market for inventory.

3. Product, Payment & Customer Analysis



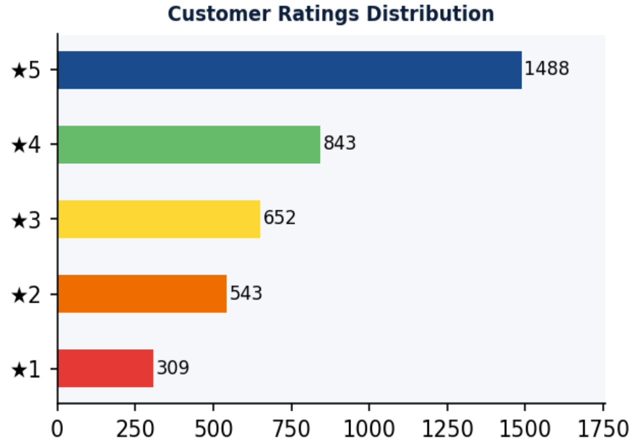


Fig 3 — UPI leads at 26.4% revenue share but all four payment methods are nearly equal, indicating highly diverse customer profiles.
Fig 4 — 38.8% of customers rated 5★; only 8.1% gave 1★ — strong overall satisfaction.

Top 5 Mobile Models by Revenue

Rank	Mobile Model	Revenue	% Share	Segment
#1	iPhone SE	■5.96 Cr	7.7%	Budget-Premium
#2	OnePlus Nord	■5.79 Cr	7.5%	Mid-Range
#3	Galaxy Note 20	■5.60 Cr	7.3%	Mid-Range
#4	Vivo Y51	■5.48 Cr	7.1%	Budget-Mid
#5	Galaxy S21	■5.33 Cr	6.9%	Premium

UPI #1 Payment
26.4% transaction share. Digital payments dominate all cities and age groups.

iPhone SE Top Model
■5.96 Cr revenue leader. Apple's budget-premium positioning wins across all segments.

4. Time-Series Trend Analysis

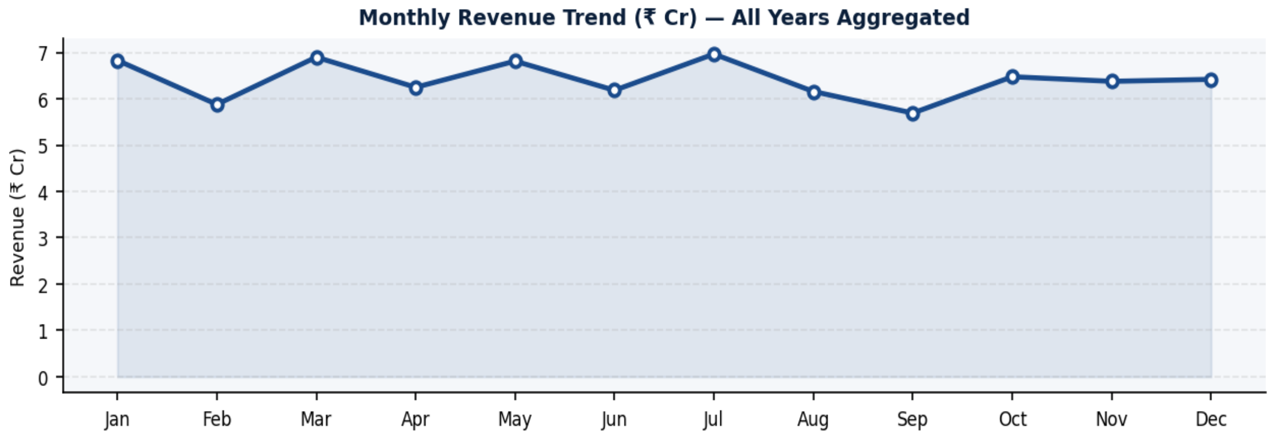


Fig 5 — July is the peak month at ■6.96 Cr. September shows a consistent dip across years. March, May, and January are the next-strongest months — suggesting Q1 promotions and mid-year launches drive demand cycles.

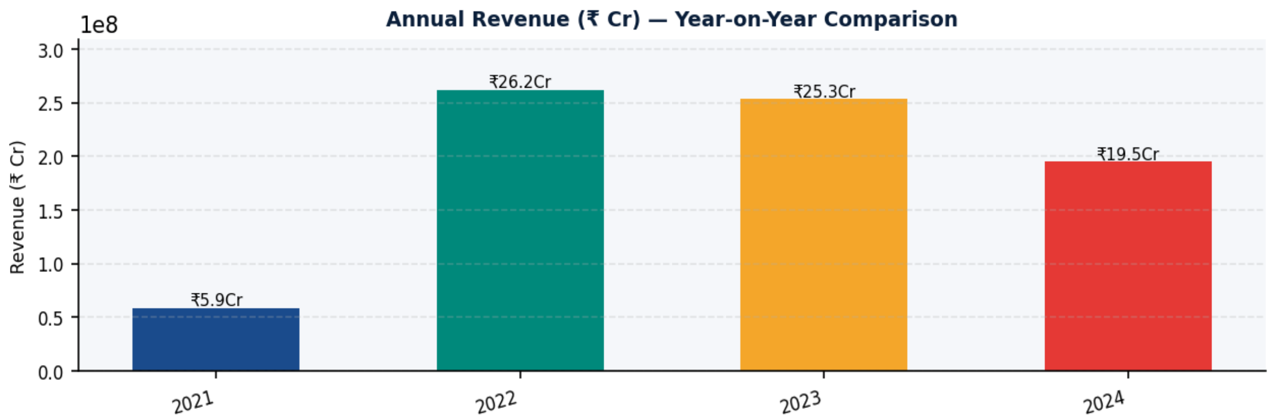


Fig 6 — 2022 was the breakout year at ■26.2 Cr (+345% YoY after partial 2021). Revenue plateaued in 2023 (–3.3%) and declined sharply in 2024 (–23%), signalling possible market saturation, increased competition, or data truncation for 2024.

5. SQL Business Queries

The following MySQL queries were written against the **mobile_sales** table to validate Power BI metrics and independently answer key business questions. Each query maps directly to a dashboard KPI or visual.

Q1 — Core KPIs: Revenue, Units, Transactions, Avg Price

```sql
SELECT
SUM(Units_Sold * Price_Per_Unit) AS Total_Revenue,
SUM(Units_Sold) AS Total_Units_Sold,
COUNT(Transaction_ID) AS Total_Transactions,
AVG(Price_Per_Unit) AS Avg_Price_Per_Unit
FROM mobile_sales;
```

→ ■76.9 Cr revenue | 19,150 units | 3,835 transactions | ■40,114 avg price

Q2 — Brand Revenue Ranking with Market Share %

```sql
SELECT Brand,
SUM(Units_Sold * Price_Per_Unit) AS Revenue,
ROUND(SUM(Units_Sold * Price_Per_Unit) /
SUM(SUM(Units_Sold * Price_Per_Unit)) OVER() * 100, 1) AS Pct_Share
FROM mobile_sales GROUP BY Brand ORDER BY Revenue DESC;
```

→ Apple ■16.2 Cr (21.0%) > Samsung ■16.0 Cr > OnePlus ■15.4 Cr

Q3 — Top Cities by Revenue & Geographic Market Share

```sql
SELECT City,
ROUND(SUM(Units_Sold * Price_Per_Unit)/1e7, 2) AS Revenue_Cr,
ROUND(SUM(Units_Sold * Price_Per_Unit) /
SUM(SUM(Units_Sold * Price_Per_Unit)) OVER() * 100, 1) AS Market_Share_Pct
FROM mobile_sales GROUP BY City
ORDER BY Revenue_Cr DESC LIMIT 5;
```

→ Delhi 26.5% | Mumbai 16.5% | Ranchi 4.0% | Chennai 4.0% | Rajkot 3.6%

Q4 — Year-on-Year Revenue Growth (Window Function)

```sql
SELECT Year,
ROUND(SUM(Units_Sold * Price_Per_Unit)/1e7, 2) AS Revenue_Cr,
ROUND((SUM(Units_Sold * Price_Per_Unit) -
LAG(SUM(Units_Sold * Price_Per_Unit)) OVER (ORDER BY Year)) /
LAG(SUM(Units_Sold * Price_Per_Unit)) OVER (ORDER BY Year) * 100, 1) AS YoY_Pct
FROM mobile_sales GROUP BY Year ORDER BY Year;
```

→ 2022: +345.4% | 2023: –3.3% | 2024: –23.0%

Q5 — Payment Method: Transactions & Revenue Share

```sql
SELECT Payment_Method,
COUNT(*) AS Transactions,
ROUND(SUM(Units_Sold * Price_Per_Unit)/1e7, 2) AS Revenue_Cr,
ROUND(COUNT(*) * 100.0 / SUM(COUNT(*)) OVER(), 1) AS Txn_Share_Pct
FROM mobile_sales GROUP BY Payment_Method ORDER BY Revenue_Cr DESC;
```

→ UPI: 1,011 txns (26.4%) | Debit: 948 | Credit: 947 | Cash: 929 — nearly equal

Q6 — Customer Satisfaction by Brand (Avg Rating + 5★ Count)

```
SELECT Brand,
ROUND(AVG(Customer_Ratings), 2) AS Avg_Rating,
COUNT(CASE WHEN Customer_Ratings = 5 THEN 1 END) AS Five_Star_Reviews
FROM mobile_sales GROUP BY Brand ORDER BY Avg_Rating DESC;
```

→ Xiaomi 3.72★ > Apple 3.71★ > Samsung 3.70★ > OnePlus 3.68★ > Vivo 3.65★

6. Business Recommendations

■ **Prioritise Delhi & Mumbai:** These two cities contribute 43% of total revenue. Dedicated stock buffers, faster SLAs, and city-specific promotions here will yield the highest ROI of any geographic investment.

■ **Investigate the 2024 Revenue Decline:** A –23% YoY decline demands urgent root-cause analysis. SQL Q4 isolates the year; the next step is drilling by city and brand to identify where the drop is concentrated.

■ **Capitalise on July Peak Demand:** July is the top month consistently across years. Pre-loading inventory in June and launching offers 2 weeks before month-end can maximise this seasonal demand window.

■ **Incentivise UPI & Card Payments:** Digital payments (UPI + cards = ~75% share) should be incentivised with cashback or loyalty points, reducing cash handling costs and improving transaction tracking quality.

■ **Double Down on iPhone SE & OnePlus Nord:** The top two revenue models. Prioritise display placement, EMI bundling, and cross-sell accessories for these two SKUs to drive outsized returns per store visit.

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